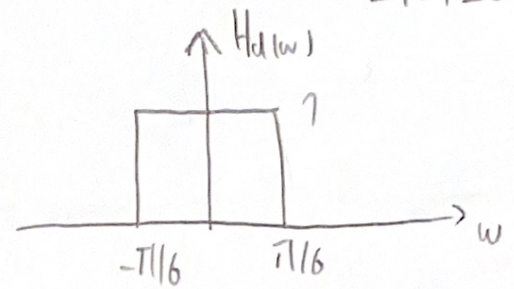


EE473 - HOMEWORK #6

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$$1) H_d(\omega) = \begin{cases} 1, & \text{for } |\omega| \leq \frac{\pi}{6} \\ 0, & \text{for } \frac{\pi}{6} < |\omega| \leq \pi \end{cases}$$



↳ This filter will have non-causal impulse response. We can multiply with a linear phase term to make it causal. $\Rightarrow$ delay = $\frac{M-1}{2} = 12$

Let us consider $H_d(\omega) = \begin{cases} e^{-j12\omega}, & |\omega| \leq \pi/6 \\ 0, & \pi/6 < |\omega| \leq \pi \end{cases}$ *magnitude is still preserved.

a)

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} e^{j\omega(n-12)} d\omega = \frac{\sin(\frac{\pi}{6}(n-12))}{\pi(n-12)}$$

We can just truncate this response with a rectangular window

$$w(n) = \begin{cases} 1, & 0 \leq n \leq 24 \\ 0, & \text{o.w.} \end{cases} \Rightarrow h(n) = h_d(n) w(n) = \begin{cases} \frac{\sin(\frac{\pi}{6}(n-12))}{\pi(n-12)}, & 0 \leq n \leq 24 \\ 0, & \text{o.w.} \end{cases}$$

b)

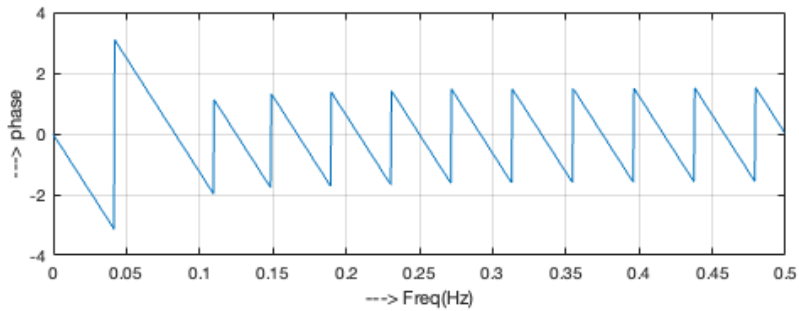
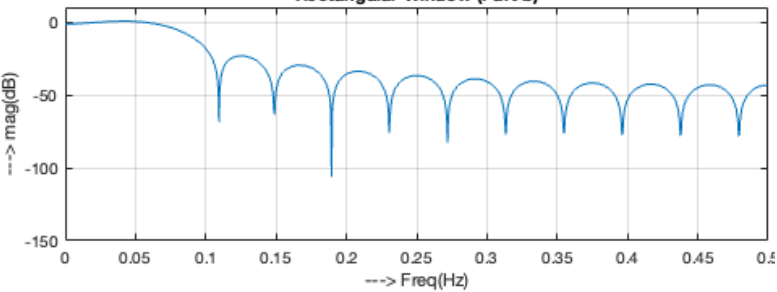
$$H(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\theta) W(\omega - \theta) d\theta \Rightarrow \text{convolution of } H_d(\omega) \text{ and } W(\omega) = \mathcal{F}\{w(n)\} \text{ in freq. domain}$$

where $W(\omega) = \sum_{n=0}^{24} e^{-j\omega n} = \frac{1 - e^{-j\omega 25}}{1 - e^{-j\omega}} = \frac{e^{-j\omega \frac{25}{2}} \sin(\frac{25\omega}{2})}{e^{-j\omega \frac{1}{2}} \sin(\frac{\omega}{2})} = e^{-j\omega 12} \frac{\sin(\frac{25\omega}{2})}{\sin(\frac{\omega}{2})}$

$$H(\omega) = \frac{e^{-j\omega 12}}{2\pi} \int_{-\pi/6}^{\pi/6} \frac{\sin(\frac{25(\omega - \theta)}{2})}{\sin(\frac{\omega - \theta}{2})} d\theta$$

$$|H(\omega)| = \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} \frac{\sin(\frac{25(\omega - \theta)}{2})}{\sin(\frac{\omega - \theta}{2})} d\theta, \quad \angle H(\omega) = \begin{cases} -12\omega, & H(\omega) > 0 \\ -12\omega + \pi, & H(\omega) < 0 \end{cases}$$

Rectangular Window (Part b)



c) Hamming window : $w(n) = 0.54 - 0.46 \cos \frac{2\pi n}{M-1}$, for $0 \leq n \leq 25$

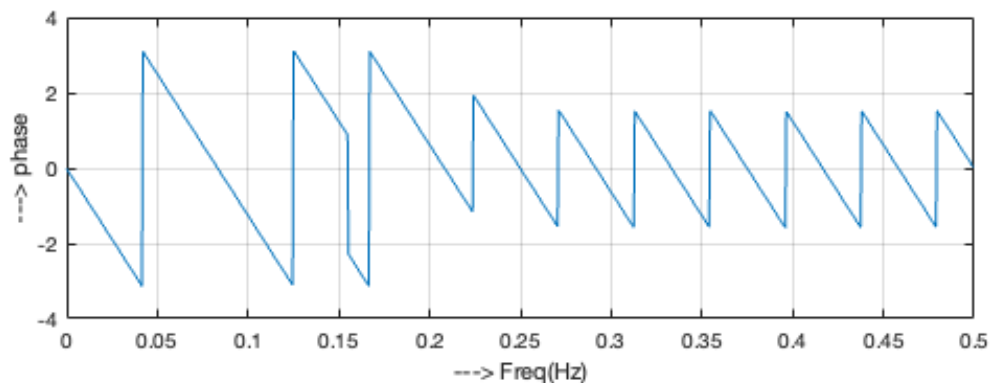
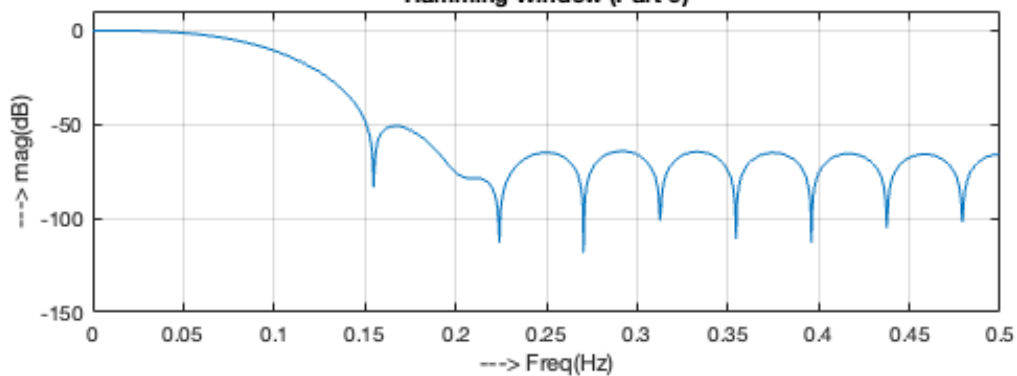
$$W(\omega) = \left\{ 0.54 \delta(\omega) - \frac{0.46}{2} \left[\delta\left(\omega - \frac{2\pi}{M-1}\right) + \delta\left(\omega + \frac{2\pi}{M-1}\right) \right] \right\}$$

$$* e^{-j\omega 12} \frac{\sin\left(\frac{25\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

$$H(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\theta) W(\omega - \theta) d\theta = \left[0.54 H_d(\omega) - 0.23 H_d\left(\omega - \frac{\pi}{12}\right) - 0.23 H_d\left(\omega + \frac{\pi}{12}\right) \right]$$

$$* e^{-j\omega 12} \frac{\sin\left(\frac{25\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

Hamming Window (Part c)



2)

$$H_r\left(\frac{2\pi k}{15}\right) = \begin{cases} 1 & k=0,1,2,3 \\ T & k=4 \\ 0 & k=5,6,\dots \end{cases}$$

→ $\{h(n)\}$ symmetric linear phase system with $M=15$

$$H(\omega) = H_r(\omega) e^{-j\omega \frac{M-1}{2}} = H_r(\omega) e^{-j\omega 7}$$

$$H\left(\frac{2\pi k}{15}\right) = H_r\left(\frac{2\pi k}{15}\right) e^{-j \frac{2\pi k}{15} \cdot 7}$$

$$h(n) = \frac{1}{15} \sum_{k=0}^{14} H\left(\frac{2\pi k}{15}\right) e^{j \frac{2\pi k}{15} n}$$

$$h(n) = \frac{1}{15} \sum_{k=0}^{14} H_r\left(\frac{2\pi k}{15}\right) e^{j \frac{2\pi k}{15} (n-7)} = \frac{1}{15} \sum_{k=0}^{14} (-1)^k H_r\left(\frac{2\pi k}{15}\right) e^{j \frac{2\pi k}{15} (n+\frac{1}{2})}$$

Recalling that $H_r\left(\frac{2\pi (M-k)}{15}\right) = H_r\left(\frac{-2\pi k}{15}\right) = H_r\left(\frac{2\pi k}{15}\right)$

$$h(n) = \frac{1}{15} \left[H_r(0) + 2 \sum_{k=1}^4 H_r\left(\frac{2\pi k}{15}\right) \cos\left[\frac{2\pi k}{15} \left(n+\frac{1}{2}\right)\right] \right]$$

$$= \frac{1}{15} \left[1 + 2 \cos \frac{2\pi}{15} \left(n+\frac{1}{2}\right) + 2 \cos \frac{2\pi}{15} 2 \left(n+\frac{1}{2}\right) + 2 \cos \frac{2\pi}{15} 3 \left(n+\frac{1}{2}\right) \right.$$

$$\left. + 2T \cos \frac{2\pi}{15} 4 \left(n+\frac{1}{2}\right) \right] \Rightarrow \text{Calculated using MATLAB}$$

For $T=0.4$

$$\begin{aligned} h(0) &= -0.0141 \\ h(1) &= -0.0019 \\ h(2) &= -0.0400 \\ h(3) &= 0.0122 \\ h(4) &= -0.0122 \end{aligned}$$

$T=0.2$

$$\begin{aligned} h(0) &= -0.0320 \\ h(1) &= 0.0196 \\ h(2) &= 0.0533 \\ h(3) &= -0.0121 \\ h(4) &= -0.0996 \end{aligned}$$

$T=0.5$

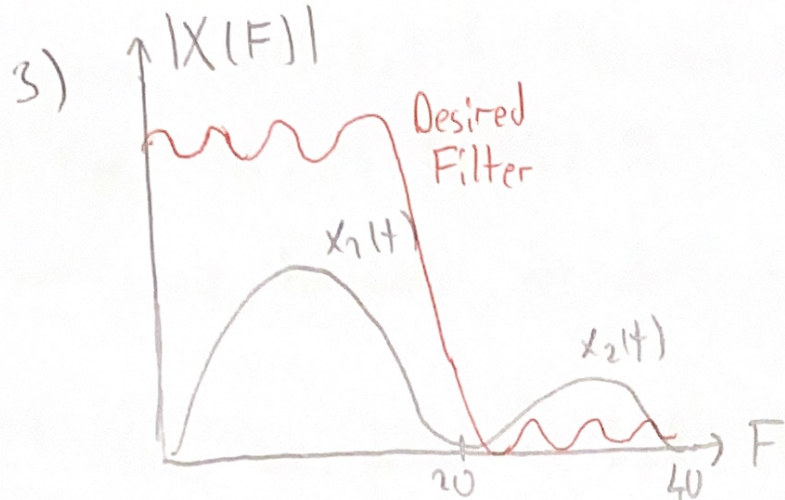
$$\begin{aligned} h(0) &= -0.0052 \\ h(1) &= -0.0127 \\ h(2) &= 0.0333 \\ h(3) &= 0.0244 \\ h(4) &= -0.0873 \end{aligned}$$

$T=0.6$

$$\begin{aligned} h(0) &= 0.0037 \\ h(1) &= -0.0235 \\ h(2) &= 0.0267 \\ h(3) &= 0.0366 \\ h(4) &= -0.0837 \end{aligned}$$

$T=0.75$

$$\begin{aligned} h(0) &= 0.0171 \\ h(1) &= -0.0397 \\ h(2) &= 0.0167 \\ h(3) &= 0.0549 \\ h(4) &= -0.0770 \end{aligned}$$



$$\delta_1 = 0.02 \quad \text{over} \quad 0 \leq |F| \leq 15 \text{ kHz}$$

$$F_p = 15 \text{ kHz}$$

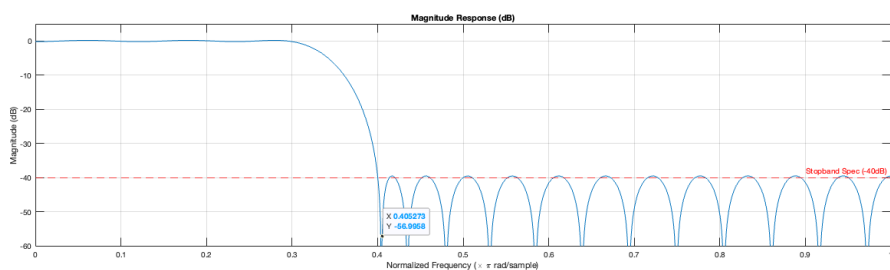
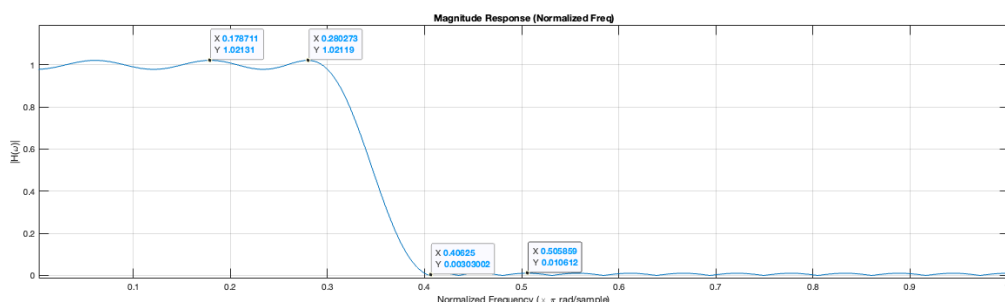
$$\delta_2 = -40 \text{ dB} = 10^{-40/20} = 0.01$$

$$F_s = 20 \text{ kHz}$$

$$B = 40 \text{ kHz}, \quad F_{\text{Nyq}} = \frac{F_s}{2} = 50 \text{ kHz}$$

Normalized freqs: $f_p = \frac{15}{50} = 0.3$, $f_s = \frac{20}{50} = 0.4$

a) Using "firls" command on MATLAB equiripple filter is found as $h_{\text{firls}} = \{0.35, -0.0065, \dots\}$ $\Rightarrow M = 37$, order (length of filter)



As seen from the figure, approximated digital filter closely matches the specifications with.

$$f_p \approx 0.28, \quad f_s = 0.406, \quad \delta_1 \approx 0.0213, \quad \delta_2 = 0.0106$$

b) Eq. 10.2.94 $\Rightarrow \tilde{M} \approx \frac{-20 \log_{10} \sqrt{\delta_1 \delta_2} - 13}{14.6 \Delta f} + 1 \approx 34$

$\hookrightarrow$ Kaiser's formula

$$\downarrow \frac{20-15}{100} = 0.05$$

Eq. 10.2.95 $\Rightarrow \tilde{M} \approx \frac{D(\delta_1, \delta_2) - f(\delta_1, \delta_2) \Delta f^2}{\Delta f} + 1 \approx 36$

$\hookrightarrow$ Hermann's formula

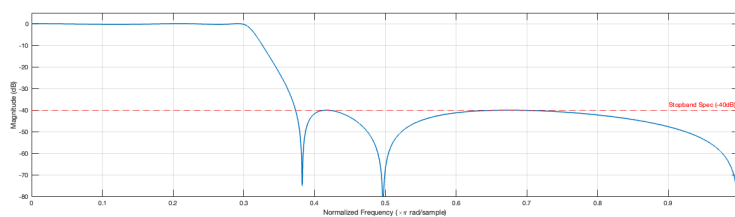
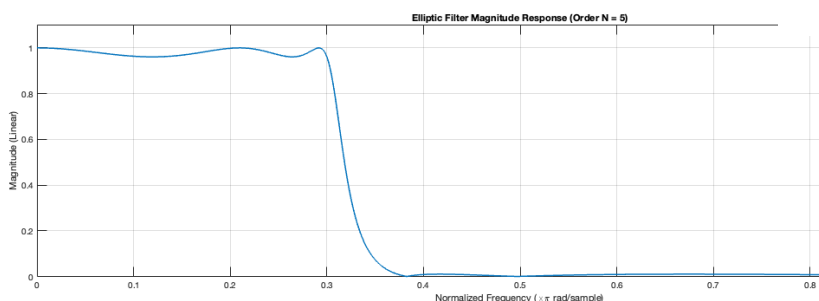
where $D(\delta_1, \delta_2)$ & $f(\delta_1, \delta_2)$ are calculated according to Eqs. 10.2.96 & 10.2.97 (Proakis et. al. DSP)

→ Clearly Hermann's formula is a better approximation

c) Now we design an IIR elliptic filter of the form

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N^2(\Omega/\Omega_p)}$$

→ MATLAB's "ellipord" function finds the minimum order elliptic filter that satisfy the desired specs.



Elliptic filter satisfies the specified conditions better than equiripple filter with specs

$$f_p = 0.292, f_s = 0.383, \delta_1 = 0.02, \delta_2 = 0.01$$

↳ ripples around 0.98

→ Elliptic filter's magnitude is 1 at maximum and it's oscillations are a little more unbalanced compared to a). Also much less ripples

d) Remez FIR

→ Linear phase FIR of order 37

→ $(M+1)/2 = 37+1/2 = 19$ multiplications

→ Storage required = 19 filter coeffs

Elliptic IIR

→ Cascade of order 5

→ $\underbrace{6}_{2^{nd} \text{ order}} + \underbrace{4}_{1^{st} \text{ order}} = 16$ multiplications

→ 16 coeffs for storage

$$4) \quad H_d(e^{j\omega}) = \begin{cases} e^{-j\omega M/2} & , \quad 0 \leq |\omega| \leq 0.3\pi \\ 0 & , \quad 0.3\pi < |\omega| < 0.6\pi \\ 0.5e^{-j\omega M/2} & , \quad 0.6\pi < |\omega| \leq \pi \end{cases}$$

$$w(n) = \begin{cases} \frac{I_0[\beta(1 - [(n-\alpha)/\alpha]^2)^{1/2}]}{I_0(\beta)} & , \quad 0 \leq n \leq M \\ 0 & , \quad \text{o.w.} \end{cases}$$

a) Desired filter has a linear phase of $-\omega \frac{M}{2}$

Since $I_0(\cdot)$ is even, $W(\omega)$ will be a real function and thus

$H(\omega) = H_d(\omega) * W(\omega)$ will have a phase of $-\omega \frac{M}{2}$

$\hookrightarrow$ For $M = 48$, delay is $\frac{M}{2} = \boxed{24}$

$$b) \quad h(n) = h_d(n) w(n)$$

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-0.3\pi}^{0.3\pi} e^{j\omega(n-24)} d\omega + 2 \times \frac{0.5}{2\pi} \int_{0.6\pi}^{\pi} e^{j\omega(n-24)} d\omega$$

$\nearrow$ since even function

$$= \frac{\sin(0.3\pi(n-24)) - 0.5 \sin(0.6\pi(n-24))}{\pi(n-24)}$$

with $h_d(24) = 0.5$

$$c) \quad B - \delta_1 \leq |H(e^{j\omega})| \leq B + \delta_1 \quad \text{in} \quad 0 \leq \omega \leq \omega_{p1}$$

$$|H(e^{j\omega})| \leq \delta_2 \quad \text{in} \quad \omega_{s1} \leq \omega \leq \omega_{s2}$$

$$C - \delta_3 \leq |H(e^{j\omega})| \leq C + \delta_3 \quad \text{in} \quad \omega_{p2} \leq \omega \leq \pi$$

$$B = 3.68 = \begin{cases} 0.11(A - 0.87) & , A > 50 \\ 0.584(A - 21)^{0.4} + 0.07886(A - 21) & , 21 \leq A \leq 50 \\ 0 & , A < 21 \end{cases}$$

If we assume $A > 50 \Rightarrow A = \frac{3.68}{0.11} + 0.87 \approx 42.09 \Rightarrow \text{contradiction}$

Assuming $21 \leq A \leq 50$: $3.68 = 0.584(A - 21)^{0.4} + 0.07886(A - 21)$

$\hookrightarrow$ By trial and error A is found as $A \approx 42.436 \text{ dB}$

$$\delta = 10^{-A/20} = 10^{-2.1218} \approx 0.00755$$

$\hookrightarrow$ Assuming equal ripples in all bands $\delta_1 = \delta_2 = \delta_3 = 0.0075$,

$\hookrightarrow$ but since $\omega_{p2} \leq |\omega| \leq \pi$ has gain 0.5 $\delta_3 = 0.003775$,

$$M = \frac{A-8}{2.285\Delta\omega} \Rightarrow \Delta\omega = \frac{34.436}{48 \times 2.285} \approx 0.3125 \text{ rad} = 0.1\pi$$

$\rightarrow$ If transition width is $\Delta\omega = 0.1\pi$, passband starts $\Delta\omega/2$ before ideal cut-off and stopband starts $\Delta\omega/2$ after cut-off

$$\omega_{p1} = 0.3\pi - 0.05\pi = 0.25\pi \text{ , , } \omega_{s1} = 0.3\pi + 0.05\pi = 0.35\pi \text{ , , } \omega_{p2} = 0.55\pi \text{ , } \omega_{s2} = 0.65\pi \text{ ,}$$

$\rightarrow$ First bands gain is $|e^{-j24\omega}| \Rightarrow B = 1$

Second bands gain is $|0.5e^{-j24\omega}| \Rightarrow C = 0.5$